Student 15 **K.Dhilip** 192365013RD 2 / 2 submitted

Q1. Food Delivery Billing System (Multiple Inheritance using Two Interfaces)

Score: 6.67 **TC: 2/3 passed**

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        double foodCost=sc.nextDouble();
        sc.nextLine();
        String couponCode=sc.nextLine();
        FoodOrder order=new FoodOrder();
        double deliveryCharge=order.calculateDeliveryCharge(foodCost);
        double discount=order.calculateDiscount(foodCost,couponCode);
        double finalBill=foodCost+deliveryCharge-discount;
        System.out.println("Food Cost : "+foodCost);
        System.out.println("Delivery Charge : "+deliveryCharge);
        System.out.println("Discount : "+discount);
        System.out.println("Final Bill : "+finalBill);

    }
}
interface DeliveryCharge{
    double calculateDeliveryCharge(double foodCost);
}
interface CouponDiscount{
    double calculateDiscount(double foodCost,String couponCode);
}
class FoodOrder implements DeliveryCharge, CouponDiscount{
    public double calculateDeliveryCharge(double foodCost){
        if(foodCost>=500){
            return 0;
        }
        else{
            return 150;
        }
    }
    public double calculateDiscount(double foodCost,String couponCode){
        if(couponCode.equals("SAVE10")){
            return foodCost*0.10;
        }
        else{
            return 0;
        }
    }
}
```

INPUT

```
600
SAVE10
```

OUTPUT

```
Food Cost : 600.0
Delivery Charge : 0.0
Discount : 60.0
Final Bill : 540.0
```

Q2. Hotel Billing System (Multiple Inheritance using One Class and One Interface)

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Main{
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        HotelBill bill=new HotelBill();
        bill.roomnumber=sc.nextInt();
        sc.nextLine();
        bill.guestname=sc.nextLine();
        bill.roomrent=sc.nextDouble();
        double serviceCharge=bill.calculateServiceCharge();
        double finalbill=bill.roomrent+serviceCharge;
        System.out.println("Room Number : "+bill.roomnumber);
        System.out.println("Guest Name : "+bill.guestname);
        System.out.println("Room Rent : "+bill.roomrent);
        System.out.println("Service Charge : "+serviceCharge);
        System.out.println("Final Bill : "+finalbill);
    }
}
class Room{
    int roomnumber;
    String guestname;
    double roomrent;
}
interface ServiceCharge{
    double calculateServiceCharge();
}
class HotelBill extends Room implements ServiceCharge{
    public double calculateServiceCharge(){
    if(roomrent>=5000){
        return roomrent*0.12;
    }
    else{
        return roomrent*0.08;
    }
}
}
```

INPUT

```
205
Dhilip
6000
```

OUTPUT

```
Room Number : 205
Guest Name : Dhilip
Room Rent : 6000.0
Service Charge : 720.0
Final Bill : 6720.0
```